# Survival Analysis Report
## Landmark Models for Early Prediction of Severely Impaired RAP

## Introduction

Continuous monitoring of cerebral physiology is essential in managing traumatic brain injury (TBI). The compensatory reserve index (RAP) reflects the brain's ability to accommodate intracranial changes, and its impairment may precede clinical deterioration. However, most studies focus on descriptive analysis rather than modeling when such deterioration occurs.

This analysis applies a time-to-event framework to model the onset of sustained severe compensatory reserve failure (i.e., RAP<0). A landmark-based design was used, where features from the early monitoring period were used to predict future deterioration among patients who remained event-free during that window.

Kaplan–Meier analysis, Cox proportional hazards models, and Random Survival Forests were employed to evaluate both interpretability and predictive performance. The objective was to determine whether early RAP and multimodal physiological features can predict the timing of severe deterioration and to compare classical and machine learning survival approaches in this setting.

## Dataset

The Winnipeg Acute TBI dataset was utilized in this analysis.

## Methods

- Landmark windows: 3 hours and 6 hours. They were chosen to balance early prediction with data reliability.
- Event detection: first sustained RAP value below -0.1 for 10 consecutive minutes. A strict threshold at 0 is sensitive to noise and small fluctuations. Using -0.1 ensures the drop is clearly below the boundary, capturing true severe deterioration rather than borderline or transient changes.
- Patients with an event during or before the landmark window were excluded from that landmark analysis.
- Patients with monitoring that did not extend beyond the landmark were also excluded.
- Features were computed from the early window only. These included RAP, ICP, CPP, MAP, and AMP summaries; separate RAP-only models also used RAP variability and burden features.
- Outcome: time from landmark to first sustained event, or time from landmark to end of follow-up if censored.
- Evaluation used log-rank tests for Kaplan-Meier analyses and concordance index (C-index) for predictive performance.

## Cohort Feasibility by Landmark

| Landmark | Included | Early-event excluded | Short follow-up excluded | Later events | Censored |
|---|---|---|---|---|---|
| 3 hours | 73 | 29 | 7 | 49 | 24 |
| 6 hours | 66 | 38 | 5 | 42 | 24 |

Interpretation: the 3-hour landmark preserved a larger analyzable cohort and more events. That generally improves model stability and partially explains why the 3-hour models were more interpretable and more competitive.

## Kaplan-Meier Results

| Grouping variable | 3-hour log-rank p | 6-hour log-rank p | Interpretation |
|---|---|---|---|
| Median split of early RAP mean | 0.0016 | 0.0283 | Significant at both landmarks; stronger separation at 3h |
| Median split of early ICP mean | 0.3531 | 0.2122 | Not significant at either landmark |

Clinical meaning: Patients with lower early RAP values experienced earlier transition into sustained severe RAP deterioration. In contrast, average early ICP alone did not separate risk groups well, suggesting that compensatory reserve status captured by RAP is more informative than ICP burden alone for this endpoint.

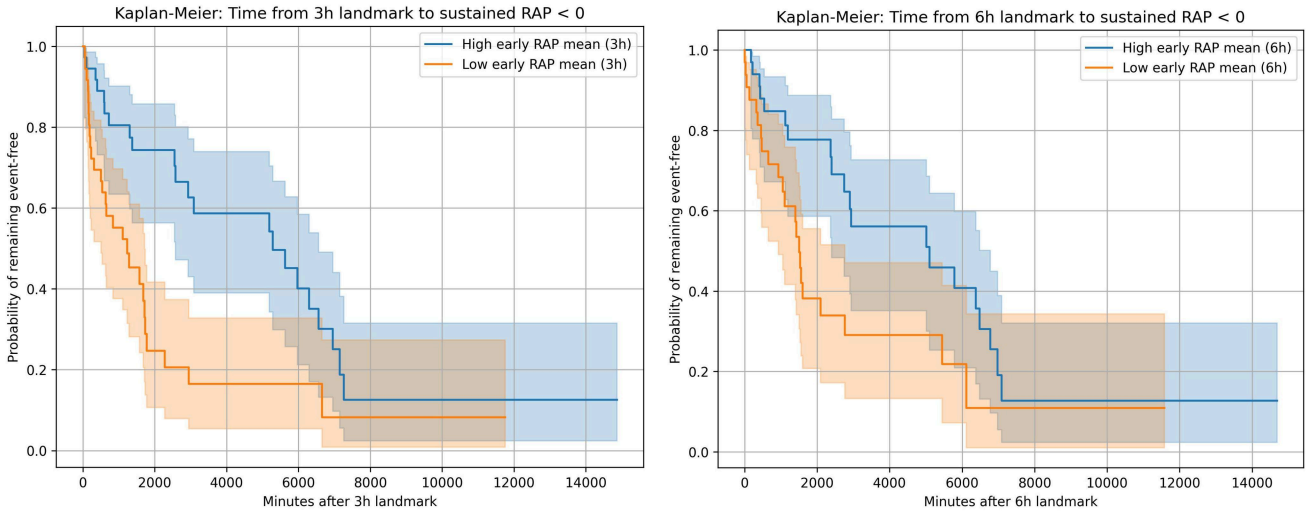


*Figure: Kaplan–Meier Survival Curves Comparing Time to Sustained RAP Deterioration by Early RAP Stratification at 3-hour and 6-hour Landmarks*

Both plots above show Kaplan–Meier curves comparing high vs low early RAP mean, but at different landmark times (3h vs 6h).

- In both cases, higher early RAP is associated with better outcomes, as the blue curves remain higher, indicating a greater probability of remaining event-free.
- The separation between groups is more pronounced at 3 hours, with earlier and clearer divergence between high and low RAP groups.
- At 6 hours, the separation is still present but weaker, suggesting some loss of discriminative signal or reduced cohort size.

- The wider confidence intervals at later times in both plots reflect increasing uncertainty due to fewer patients at risk.

Overall, the 3-hour landmark provides stronger early risk stratification, while the 6-hour model shows similar trends but with reduced clarity.

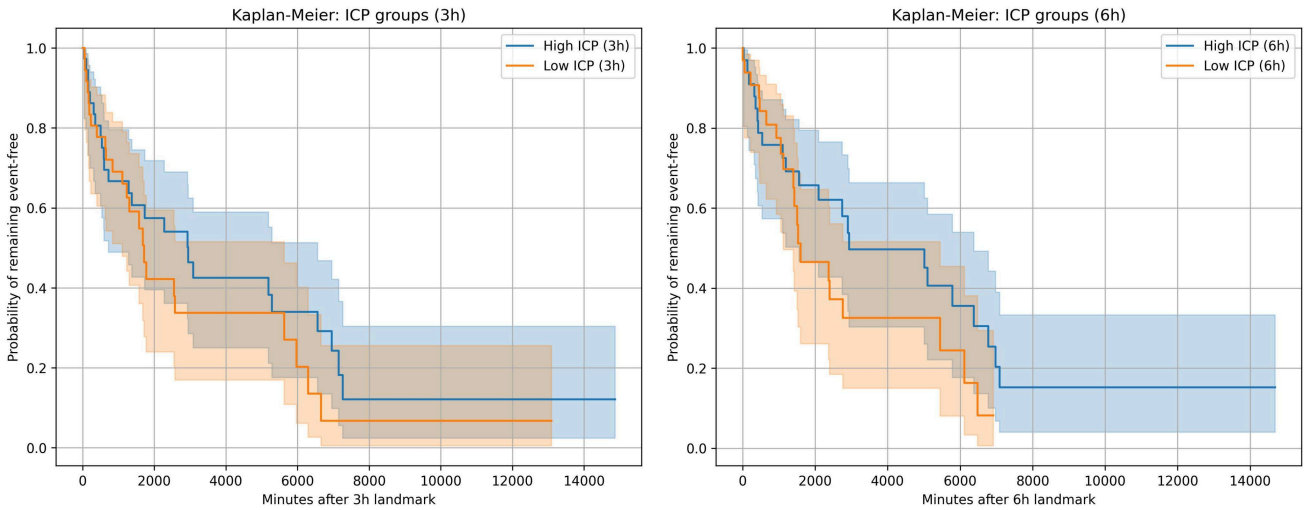


*Figure: Kaplan–Meier Survival Curves Comparing Time to Sustained RAP Deterioration by Early ICP Groups at 3-hour and 6-hour Landmarks*

In the previous figure, patients were stratified based on early RAP values, while in this figure, stratification is based on early ICP values at both 3-hour and 6-hour landmarks.

For the RAP-based curves, there is a clear separation between the groups. Patients with higher early RAP maintain a higher probability of remaining event-free, while those with lower RAP deteriorate earlier. This separation is especially pronounced at the 3-hour landmark and remains visible at 6 hours.

In contrast, the ICP-based curves show much weaker separation. The high and low ICP groups follow very similar trajectories, with considerable overlap between the confidence intervals. This indicates that early ICP alone does not effectively distinguish between patients who deteriorate earlier versus later.

Comparing the two figures, RAP provides a stronger and more consistent stratification of risk than ICP. This suggests that compensatory reserve, as captured by RAP, is more informative for predicting the timing of deterioration than pressure-based measures alone.

## Cox Proportional Hazards Models

### Model summary table

| Model | n | Events | Key significant findings | PH assumption | Convergence | C-index |
|---|---|---|---|---|---|---|
| 3h, all physiologic means | 70 | 47 | rap_mean_3h HR 0.028, p = 0.00085; | Looks okay | No warning | 0.687 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | others not significant | | | |
| 3h, RAP-only | 73 | 49 | rap_mean_3h HR 0.054, p = 0.0226 | Violations for rap_mean_3h and pct_rap_gt_0_4_3h | No warning | 0.673 |
| 6h, all physiologic means | 64 | 41 | icp_mean_6h, cpp_mean_6h, map_mean_6h significant; rap_mean_6h not significant | Severe violations | Convergence warning | 0.531 |
| 6h, RAP-only | 66 | 42 | rap_mean_6h HR 0.045, p = 0.0216 | Violation for pct_rap_gt_0_4_6h | No warning | 0.649 |

### Interpretation of Cox findings

- The most consistent signal in the Cox analyses was early RAP mean. Hazard ratios far below 1 at both landmarks indicate that higher early RAP was protective, meaning patients with better early compensatory reserve tended to deteriorate later.
- The 3-hour multivariable Cox model is the most credible Cox model in this analysis. It had the best Cox discrimination, a strong and statistically significant RAP effect, and no major proportional hazards problems.
- The 6-hour multivariable Cox model should not be treated as clinically interpretable in its current form. It produced a convergence warning and showed extreme proportional hazards violations. More importantly, ICP, CPP, and MAP were entered together even though CPP is derived from MAP and ICP, which creates structural collinearity and can distort effect directions and significance.
- The RAP-only Cox models were cleaner. They preserved the central biological message that lower early RAP predicts earlier severe deterioration, although some proportional hazards issues remained for burden variables.

## Random Survival Forest Results

| Model | Test C-index | Top positive importance features | Negative or unstable importances | Take-home |
|---|---|---|---|---|
| 3h, all physiologic means | 0.613 | rap_mean_3h, cpp_mean_3h, map_mean_3h | icp_mean_3h, amp_mean_3h | Moderate performance; RAP still leads |
| 3h, RAP-only | 0.696 | pct_rap_0_to_0_4_3h, rap_mean_3h, pct_rap_gt_0_4_3h | rap_std_3h | Best apparent test performance |
| 6h, all physiologic means | 0.669 | rap_mean_6h, map_mean_6h, amp_mean_6h | icp_mean_6h | Best 6h model; suggests nonlinear value in longer window |

| | | | | |
|---|---|---|---|---|
| 6h, RAP-only | 0.615 | rap_mean_6h, pct_rap_gt_0_4_6h | pct_rap_0_to_0_4_6h, rap_std_6h | Weaker than 6h all-variable RSF |

Interpretation: RAP remained the dominant variable in the RSF models as well. The strong 3-hour RAP-only RSF suggests that non-linear relationships within RAP-state burden may contain useful signal. However, permutation importances that become negative usually indicate instability, noise, or strong redundancy between predictors. Because these RSF results came from a single train-test split on a small cohort, they should be viewed as preliminary rather than definitive.

## Clinical Interpretation

- Early RAP appears to be a better marker of near-term severe reserve failure than mean ICP alone. This is clinically plausible because RAP is intended to reflect compensatory reserve status, whereas ICP alone may remain relatively similar across patients before overt decompensation appears.
- The persistence of the RAP signal across both landmarks supports the idea that early reserve impairment precedes later severe deterioration. In practical terms, patients entering the monitoring period with lower RAP may merit closer observation or earlier intervention.
- The 3-hour results suggest that clinically useful prognostic information is already present early in the monitoring period. This is valuable because an earlier landmark offers more time for risk stratification and potential treatment adaptation.
- The RSF finding that time spent in the borderline RAP range (0 to 0.4) contributed strongly at 3 hours suggests that transitional physiology may matter. Patients who spend more time near the boundary between adequate and impaired reserve may be unstable even if their mean RAP is not frankly low.

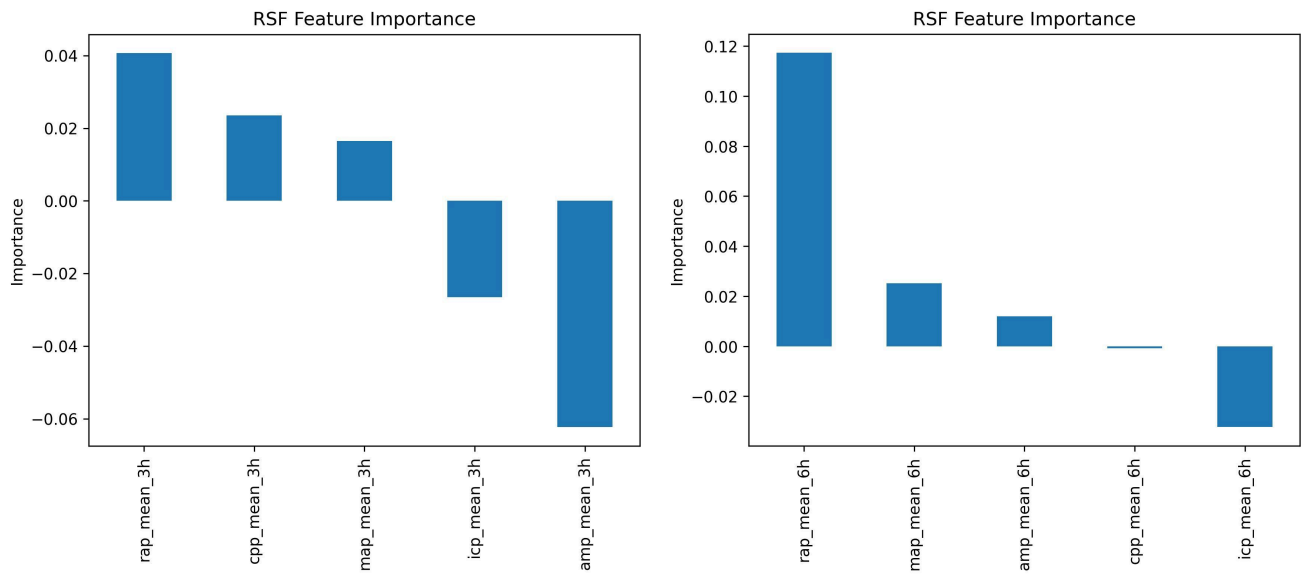


*Figure: Random Survival Forest Feature Importance for Early Physiological Predictors at 3-hour and 6-hour Landmarks*

This figure shows the relative importance of early physiological features in predicting time to sustained RAP deterioration using Random Survival Forest models at 3-hour and 6-hour landmarks.

Across both models, early RAP mean is the most influential predictor, with a stronger contribution at 6 hours. Other variables such as MAP and CPP show moderate importance, while ICP and AMP contribute less or show negative importance, suggesting limited or unstable predictive value.

## Model Comparison and Preferred Framing

- For interpretability and statistical defensibility, the preferred primary model is the 3-hour Cox model with carefully selected predictors, especially RAP-based measures.
- For predictive performance, the strongest observed result was the 3-hour RAP-only RSF (test C-index 0.696), but that advantage over the 3-hour Cox model was small and needs confirmation with repeated resampling or cross-validation.
- The 6-hour landmark is not useless, but it is less attractive as the main analysis because it excludes more early deteriorators and leaves fewer events for modeling. It may still be useful as a sensitivity analysis.
- Taken together, these findings support a simple story: an earlier landmark is more efficient, and RAP is the most clinically meaningful predictor family for this endpoint.

## Conclusion

This analysis demonstrates that early RAP-derived features provide meaningful predictive information regarding the timing of severe compensatory reserve deterioration. Across all approaches, lower early RAP was consistently associated with earlier deterioration.

Among the tested configurations, the 3-hour landmark offered the best balance between sample retention, event yield, interpretability, and predictive performance. Both Kaplan–Meier and Cox models showed clear and consistent stratification based on RAP, while ICP-based stratification demonstrated limited discriminative value.

Random Survival Forest models further supported the dominant role of RAP, although their performance advantage over Cox models was modest. Overall, these findings suggest that early compensatory reserve status, as captured by RAP, is more informative than pressure-based measures alone for anticipating near-term deterioration.